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Effect of creatine supplementation on kidney function: a systematic review and meta-analysis

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Abstract

Background Creatine monohydrate is a widely used dietary supplement with proven benefits in athletic performance and potential therapeutic applications in clinical populations. However, concerns regarding its impact on renal function persist, largely due to elevated serum creatinine levels associated with creatine intake.

Methods A comprehensive literature search was conducted in PubMed, Scopus, Web of Science, and Google Scholar for studies published between January 2000 and March 2025. Primary outcomes were serum creatinine and glomerular filtration rate (GFR). Meta-analyses were performed using a random-effects model, with subgroup analysis by supplementation duration.

Results Twenty-one studies were included in the systematic review, and 12 studies (177 participants in the creatine group and 263 in control) were eligible for meta-analysis of serum creatinine levels. Creatine supplementation was associated with a small but statistically significant increase in serum creatinine (MD: 0.07 µmol/L; 95% CI: 0.01 to 0.12; $p=0.03$). Subgroup analysis of the creatinine outcome based on follow-up duration revealed that the intervention effect was significant in studies with a follow-up of ≤ 1 week (MD = 0.12; 95% CI: 0.03 to 0.21). However, no significant effect was observed in studies with a follow-up of 1 to 12 weeks (MD = 0.04; 95% CI: −0.09 to 0.17). The effect became significant again in studies with a follow-up of more than 12 weeks. The meta-analysis revealed a non-statistically significant differences in GFR following creatine supplementation compared to control.

Conclusion Creatine supplementation is associated with a modest, transient increase in serum creatinine levels, likely due to metabolic turnover rather than renal impairment. No significant changes were observed in GFR, suggesting preserved kidney function.

Clinical trial number Not applicable.

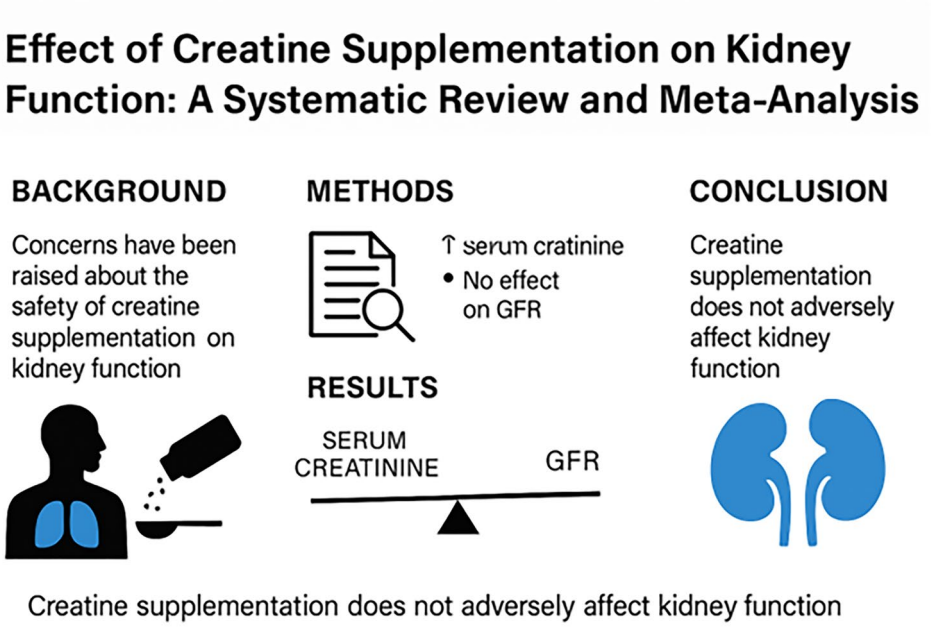
Keywords Creatine supplementation, Kidney function, Serum creatinine, Glomerular filtration rate, Renal safety, Meta-analysis

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Graphical Abstract



Introduction

To improve physical performance and overall health, both competitive and recreational athletes are increasingly turning to methods that provide a performance advantage. Among these methods, nutritional and exercise approaches are commonly employed to enhance training results [1]. Creatine monohydrate, often simply called creatine (α-methyl-guanidine-acetic acid: $C_4H_9N_3O_2$), has become one of the most thoroughly studied and commonly used dietary supplements for enhancing physical performance [2, 3]. Through a process involving amino acids glycine and arginine, the liver, kidneys, and pancreas create naturally occurring creatine. It plays a crucial role in supporting cellular energy metabolism [2]. Its ability to raise intramuscular phosphocreatine concentrations helps to rapidly resynthesize adenosine triphosphate (ATP) during prolonged exercise [4]. Beyond its ergogenic advantages, increasing data indicate that creatine may have therapeutic effects in several clinical settings, including rheumatic illnesses [5, 6], and neurodegenerative disorders [7, 8], highlighting its broader applicability beyond the athletic population.

A fundamental assessment of kidney performance, glomerular filtration rate (GFR) is usually deduced from solute removal. Exogenous markers such as inulin are used in the most accurate approach, mGFR, but this method is usually limited to specific settings. eGFR is often calculated using serum biomarkers such as creatinine (Crn), a byproduct of creatinine metabolism [1, 2]. Blood Crn levels may rise as a result of chronic creatine supplementation, which raises total body creatine. This increase, meanwhile, does not always indicate compromised renal function. Because of this, estimating GFR in creatine supplementation trials using only creatinine-based metrics can be deceptive and is mainly insufficient for precisely evaluating renal health in this situation [3, 4].

There are still concerns about the safety of creatine supplementation, particularly in relation to kidney function, although many studies have not observed any significant changes in important renal markers, including mGFR, cystatin C, proteinuria, or albuminuria. Early reports are the main source of these concerns [5–10], animal studies [11], and the clinical reliance on serum creatinine. However, many studies have shown that oral creatine use did not lead to kidney damages [12–14]. The goal of the current study is to conduct a thorough and updated systematic review because of the contradictory findings surrounding this topic, the critical importance of comprehending its implications.

Methods

The present systematic review and meta-analysis was conducted in accordance with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. Each step was performed by two independent reviewers and any discrepancies between reviewers were resolved through discussion or adjudication by a third reviewer.

Search strategy

A comprehensive literature search was performed across multiple electronic databases including PubMed, Scopus, Web of Science, and Google scholar to identify relevant studies investigating the effects of creatine supplementation on kidney function in human subjects from January 2000 to March 2025. The main search terms included:

("creatine supplementation" OR "creatine monohydrate" OR "creatine intake")

AND ("kidney function" OR "renal function" OR "glomerular filtration rate" OR "serum creatinine" OR "cystatin C" OR "BUN" OR "renal safety" OR "renal impairment" OR "kidney toxicity").

Boolean operators (AND/OR), and database-specific filters were applied where appropriate. Language was restricted to English, and only studies involving human participants were included.

In addition to the electronic search, manual screening of reference lists from included articles and relevant reviews was performed to identify any additional eligible studies not captured through database searches. Duplicates were removed using EndNote, and the screening process was carried out independently by two reviewers in two stages: [1] title and abstract screening, and [2] full-text review.

Eligibility criteria

Inclusion criteria included observational cohort or case-control studies, non-randomized trials, RCTs, and pre-post experimental designs involving human subjects of any age, sex, or health status (e.g., healthy individuals, athletes, or patients with chronic conditions). Regardless of dosage, duration, or particular form (e.g., creatine monohydrate), oral creatine supplementation was used in all included investigations. Serum creatinine, glomerular filtration rate (GFR), and descriptive results pertaining to renal function were the main outcomes taken from each trial.

Articles published prior to 2000 were not included, nor were non-human studies (such as animal or in vitro models), case reports, reviews, editorials, commentaries, or conference abstracts with incomplete data, studies with no renal outcomes or with insufficient data for extraction, duplicate publications, or studies with overlapping data.

Data extraction

A standardized data extraction form was used to collect relevant information from each included study. Two reviewers independently extracted the following data: authors, year of publication, country, number of participants, mean age, health status, dose, frequency, duration, loading/maintenance phases, main findings related to renal function (e.g., serum creatinine, GFR), and any kidney-related complications or safety concerns.

Risk of bias assessment

Randomized controlled trials (RCTs) were evaluated using the Cochrane RoB-2 domains. Observational and pre-post studies were evaluated with the Newcastle–Ottawa Scale (NOS) and simplified checklists.

Statistical analysis

In this study, data on two primary outcomes serum creatinine and GFR were extracted from each eligible study, including the mean, standard deviation, and sample size for both intervention and control groups. The mean difference (MD) between groups was calculated as the primary effect size. To synthesize the data, heterogeneity across studies was assessed using the I^2 statistic, Cochran's Q test, and the Galbraith plot. A random-effects model was applied when substantial heterogeneity was detected ($I^2 > 50\%$ and $p < 0.05$); otherwise, a fixed-effects model was used. For the creatinine outcome, a subgroup analysis was conducted based on duration of follow-up. Potential publication bias was evaluated using funnel plots and Egger's regression test. All statistical analyses were conducted using STATA version 17.

Results

Search and study selection

The study selection process is summarized in Fig. 1. An initial set of records was identified through database searching and manual reference checks. After removing duplicates, screening titles and abstracts, and assessing full texts against the predefined inclusion and exclusion criteria, a total of 21 studies were deemed eligible [15–35]. Of the 21 studies, 12 were included in the meta-analysis as they provided the necessary quantitative data [16, 17, 19, 21–23, 25–27, 29, 31, 35]. Among these, two studies featured two intervention arms [17, 31], and one study included three arms [23]. In total, 16 sets of results were considered for meta-analysis. The study by Solis et al. [34] was excluded from the meta-analysis due to its crossover design, which was not compatible with the pooling criteria. The characteristics and summary of extracted data is presented in Table 1.

Creatinine level

Meta-analysis

A total of 14 studies evaluating serum creatinine levels before and after intervention were included in the meta-analysis. The intervention (creatine supplementation) group comprised 177 participants, while the control group included 263 participants. Figure 2 shows the pooled effects of creatine supplementation on serum creatinine levels in both groups. The meta-analysis revealed

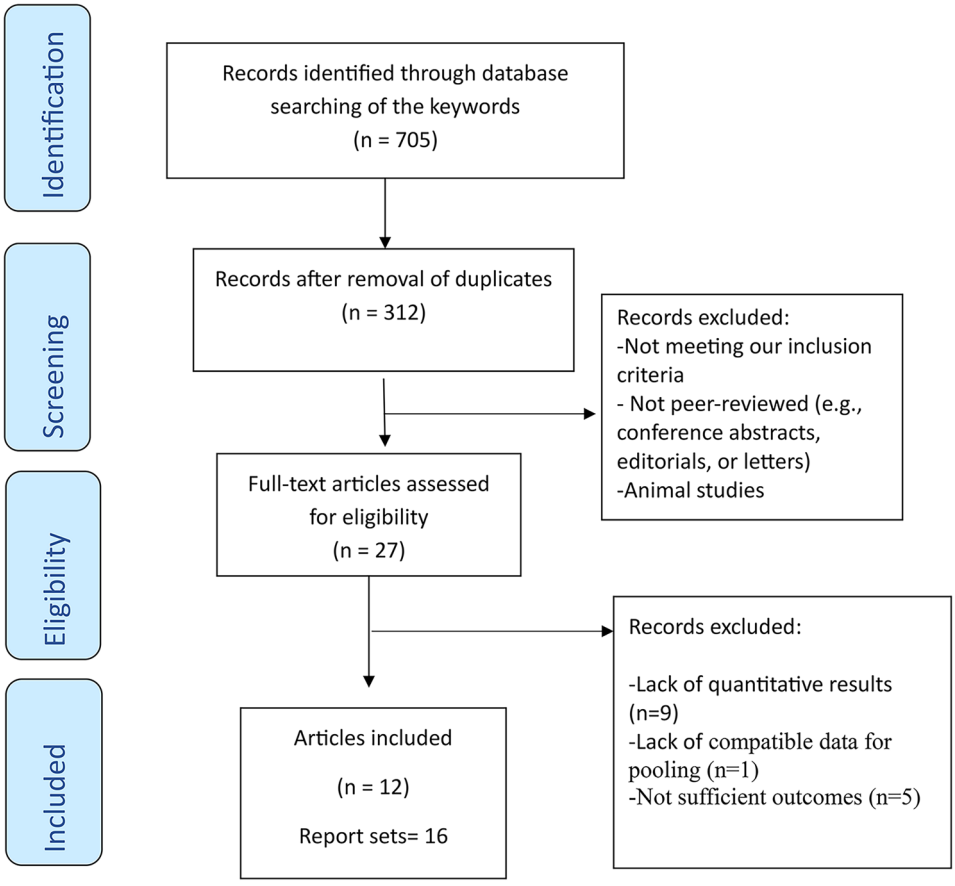


Fig. 1 Flowchart of study selection

a statistically significant increase in serum creatinine levels following creatine supplementation compared to control ($p = 0.03$), with a MD = 0.07; 95% CI: (0.01,0.12).

Subgroup analysis

Figure 3 shows the results of subgroup analysis of the creatinine outcome based on follow-up duration. Our pooled results revealed that the intervention effect was significant in studies with a follow-up of ≤ 1 week (MD = 0.12; 95% CI: 0.03 to 0.21). However, no significant effect was observed in studies with a follow-up of 1 to 12 weeks (MD = 0.04; 95% CI: − 0.09 to 0.17). The effect became significant again in studies with a follow-up of more than 12 weeks, suggesting a possible delayed but meaningful elevation in serum creatinine over extended supplementation periods.

Assessment of publication bias for creatinine level

Potential publication bias was assessed using both a regression-based funnel plot (Fig. 4) and a classic funnel plot (Fig. 5). Visual asymmetry was observed in both plots, suggesting possible small-study effects or reporting bias particularly a lack of studies showing null or negative effects. The regression funnel plot showed a slight positive slope, while the classic funnel plot indicated an uneven distribution of studies around the mean. These findings may be influenced by publication bias or study heterogeneity. However, trim-and-fill analysis showed no change in the pooled results, indicating that the overall conclusions are robust despite the observed asymmetry.

Heterogeneity

The meta-analysis revealed substantial heterogeneity for serum creatinine outcomes (I^2 = 69.73%), indicating considerable variability across studies, likely due to differences in study populations, creatine dosing, and follow-up durations. In contrast, heterogeneity for GFR outcomes was moderate (I^2 = 33.17%), suggesting that the variation in effect sizes was relatively low. These findings justified the use of a random-effects model for both outcomes to account for between-study variability and ensure more conservative pooled estimates.

GFR

Meta-analysis

A total of 5 studies evaluating GFR before and after intervention were included in the meta-analysis. These studies compared GFR after creatine use with placebo or control

Table 1 Summary of data extraction of the included studies on effect of creatine supplementation on kidney function

Authors/year Country	Creatine sample size/ mean age	Control sample size/ mean age	Subjects' status	Creatine group	Control group	Follow-up time	Outcomes
Gualano et al. 2008 [22], Brazil	9 24.2 ± 5.0	9 24.6 ± 4.2	Healthy sedentary males	0.3 g daily per kg of body weight for the first week, and 0.15 g for the next 11 weeks	Placebo (dextrose)	4 weeks & 12 weeks	no serious adverse effects on kidney function
Mayhew et al. 2002 [26], USA	10 20.5 ± 1.4	13 20.1 ± 0.8	Football Players	5 and 20 g daily (13.9 ± 5.8 g)	-	0.25 to 5.6 years (mean ± SD = 2.9 ± 1.8 years): 132 weeks	no long-term serious adverse effects on kidney or liver functions
Carvalho et al. 2011, Brazil [17]	12 23.0 ± 3.2 & 11 25.2 ± 7.4	12 24.3 ± 4.9	Healthy adults	Loading: creatine 20 g/d for 7 days Group1: 0.03 g/kg body weight once daily after training. Group 2: Fixed dose of 5 g creatine taken once daily after training.	placebo, maltodextrin (MIDWAY). Loading: 20 g/d for 7 days Maintenance: 9.7 ± 5.7 g/d	8 weeks	no serious adverse effects on kidney and liver functions in both creatine groups with different doses
Mihic et al. 2000, Canada [27]	15	15	Healthy, normotensive subjects	20 g/day of creatine, divided into five doses	20 g/day of placebo	5 days	no serious adverse effects on kidney function and blood pressure
Robinson et al. 2000, UK [31]	Group 1 = 7 Group 2 = 6	Group 1 = 7 Group 2 = 6	Healthy adults with no exercise	Cr 5 g + 1 g glucose dissolved in a warm drink four times a day for five days.	500 ml of a carbohydrate solution four times a day for five days	Group 1: 6 days & Group 2: 5 days + 6 weeks	no serious adverse effects on kidney, liver, or hematological functions
Volek et al. 2001, USA [35]	10 23.0 ± 1.0	10 23.1 ± 1.0	Healthy adults with heat exercise 30 min of continuous cycling: 15 min at 70% of VO_2peak followed by 15 min at 60% of VO_2peak,	0.3 g/kg/d for 7 days	Placebo	7 days	no serious adverse effects on kidney
Armentano et al. 2007, USA [16]	17	18	Healthy, active duty, U.S. Army volunteers	20 g/day	Placebo	7 days	no serious long-term adverse effects on kidney
Neves et al. 2011, Brazil [29]	13	13	Postmenopausal women	creatine (20 g·day(-1) for 1 week and 5 g·day(-1) thereafter	Placebo	12 weeks	no serious adverse effects on kidney in postmenopausal women
Lugaresi et al. 2013, Brazil [25]	12	14	Young healthy males	20 g/d for 5 d followed by 5 g/d	Placebo	12 weeks	no serious adverse effects on kidney
Gualano et al. 2011, Brazil [21]	13 57.5 (5.0)	12 56.4 (8.2)	Men and women (older than 45 years) with type 2 diabetes,	5 g/day of CR monohydrate	Placebo	12 weeks	no serious adverse effects on kidney

Table 1 (continued)

Authors/year Country	Creatine sample size/ mean age	Control sample size/ mean age	Subjects' status	Creatine group	Control group	Follow-up time	Outcomes
Domingues et al. 2020, Brazil [19]	14	15	Healthy adults	:20 g/day for 1 week divided into 4 equal doses single daily doses of 5 g in the subsequent 7 weeks	Placebo	8 weeks	no serious adverse effects on kidney
Kreider et al. 2003. USA [23]	Group 1 = 12 Group 2 = 25 Group 2 = 17	44	Football players	0.1 g/ kg/day of creatine monohydrate	placebo (dextrose)	Group 1 = creatine for 0–6 months (mean 4.4 ± 1.8 months) Group 2 = creatine for 7–12 months (mean 9.3 ± 2.0 months) Group 3 = creatine for 12–21 months (mean 19.3 ± 2.4 months)	no serious long-term adverse effects on kidney

GFR: glomerular filtration rate

conditions. The intervention (creatine supplementation) group comprised 69 participants, while the control group included 74 participants. Figure 6 illustrates the pooled effects of creatine supplementation on GFR in both groups. The meta-analysis revealed a non-statistically significant differences in GFR following creatine supplementation compared to control.

Assessment of publication bias for GFR

Potential publication bias was examined using both a regression-based funnel plot (Fig. 7) and a classic funnel plot based on mean differences (Fig. 8). Visual inspection of both plots revealed asymmetry and the upward slope of the regression line point, suggesting the presence of small-study effects or publication bias. To further confirm and quantify the extent of any potential bias a trim-and-fill analysis was conducted and the results remained unchanged.

Risk of bias assessment

Table 2 summarizes the detailed risk of bias assessment for each included study, based on study design. Overall, the risk of bias in the included studies was assessed as mostly at the level of 'some concern'. While most randomized controlled trials were described as randomized, double-blind and placebo-controlled, methodological details such as sequence generation, allocation concealment, reporting of dropouts and trial registration were often missing, leading to reduced confidence. Small sample sizes and short duration of follow-up also limited several trials. Observational and before-and-after studies inherently have a higher risk of bias due to confounders, lack of control groups and possible selection bias. Overall, the evidence base provides reassuring safety data, but the overall certainty is tempered by methodological limitations, incomplete reporting, and a predominance of small, single-center trials.

Discussion

A previous systematic review and meta-analysis included 5 studies (comprising 8 outcome measures) with a total of 220 participants to evaluate serum creatinine levels, and 3 studies involving 136 participants to assess glomerular filtration rate (GFR) [36]. The present systematic review and meta-analysis included a larger body of evidence, incorporating 21 studies in total of which 12 (comprising 14 outcome measures) were eligible for quantitative synthesis of serum creatinine outcomes and 5 for GFR. This expanded dataset provides more comprehensive insights into the renal safety of creatine supplementation across diverse populations, supplementation protocols, and durations. GFR is considered the best overall indicator of renal function in both healthy individuals and patients with kidney disease. Current clinical guidelines

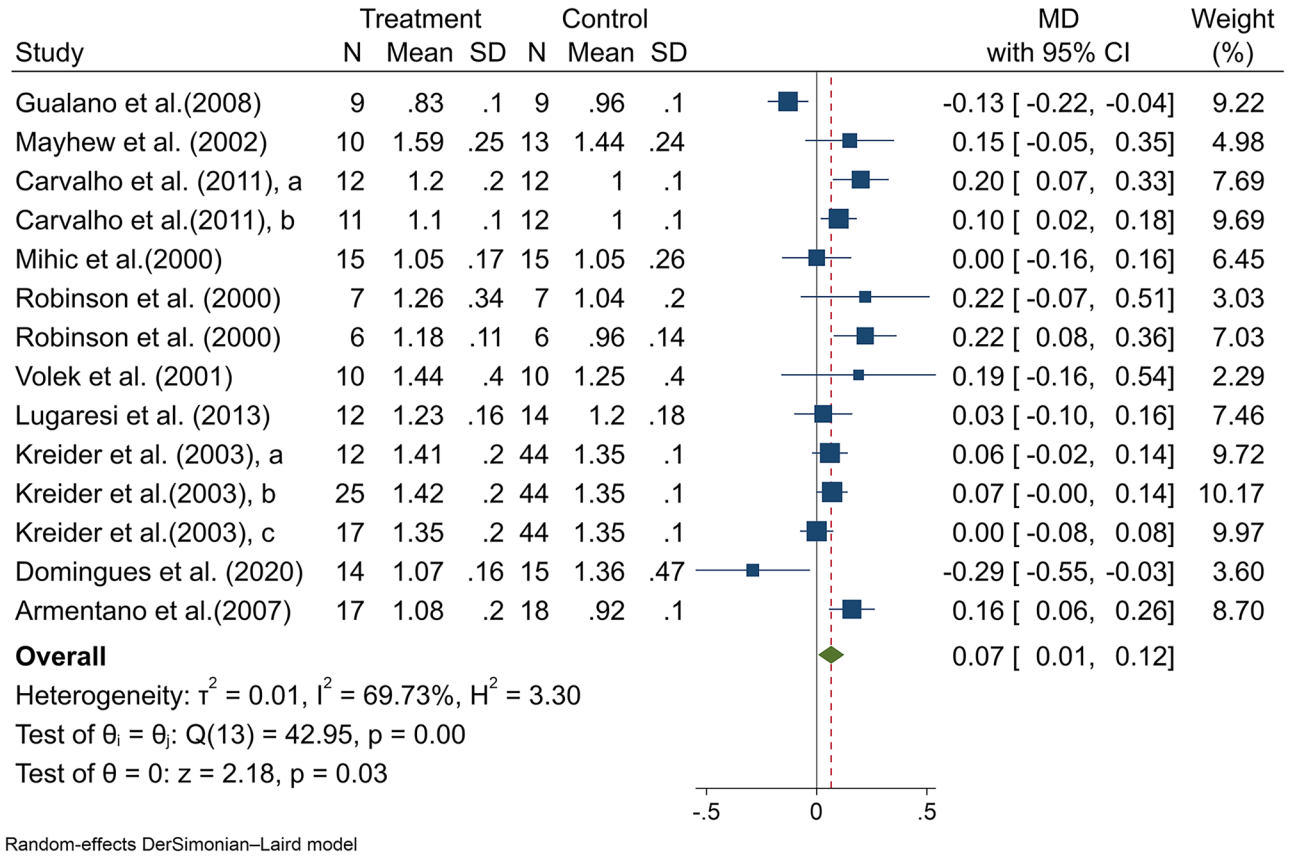


Fig. 2 Effects of creatine supplementation on serum creatinine levels

highlight that GFR reflects the integrated function of all nephrons and is therefore a more reliable index of renal health than isolated blood or urine markers. Stable GFR values are clinically more important than transient elevations in serum creatinine or other biomarkers, which may be influenced by extrarenal factors such as muscle mass, diet, or supplementation [1, 2].

Pooled results of the present meta-analysis revealed a small but statistically significant increase in serum creatinine levels following creatine supplementation. Subgroup analysis by duration of supplementation suggested that short-term (< 1 week) and long-term (> 12 weeks) interventions significantly affected serum creatinine levels. However, supplementation lasting between 1 and 12 weeks was associated with a non-significant increase.

This pattern indicates that intramuscular creatine and its metabolic byproduct, Crn, may rise quickly during the short-term phase (less than one week) of creatine supplementation. Instead of being a sign of renal failure, this early rise is usually pharmacokinetic in nature, a predictable and quantifiable biochemical reaction. This short time frame mostly records the first rise in creatinine levels prior to the onset of physiological adaptation.

The mid-term interval (1–12 weeks) indicates a transitional phase during which the body begins to adjust to increasing creatine intake. Creatine levels start to stabilize, but the system has not yet fully adapted, leading to potential fluctuations in creatinine production. Muscle creatine stores become saturated, creatine metabolism stabilizes, and renal clearance mechanisms may adjust to manage the sustained creatine load. As a result, a small but consistent elevation in serum creatinine may persist and become statistically detectable in longer-duration studies, particularly those with larger sample sizes and lower variability. Importantly, this increase appears to be statistically significant but physiologically benign, reflecting sustained creatine turnover rather than impaired kidney function. Since serum Crn is a metabolite of creatine, its elevation is a predictable physiological response rather than a sign of kidney damage [8, 37]. Therefore, relying solely on serum Crn as a marker of renal function in creatine supplementation studies may be misleading. Supporting this interpretation, the pooled analysis of GFR data from five studies demonstrated no significant changes in either estimated or directly measured GFR following creatine supplementation. This further supports the idea that reported increases in serum creatinine are not symptomatic of diminished renal function [15, 16, 27–29, 31, 34, 35].

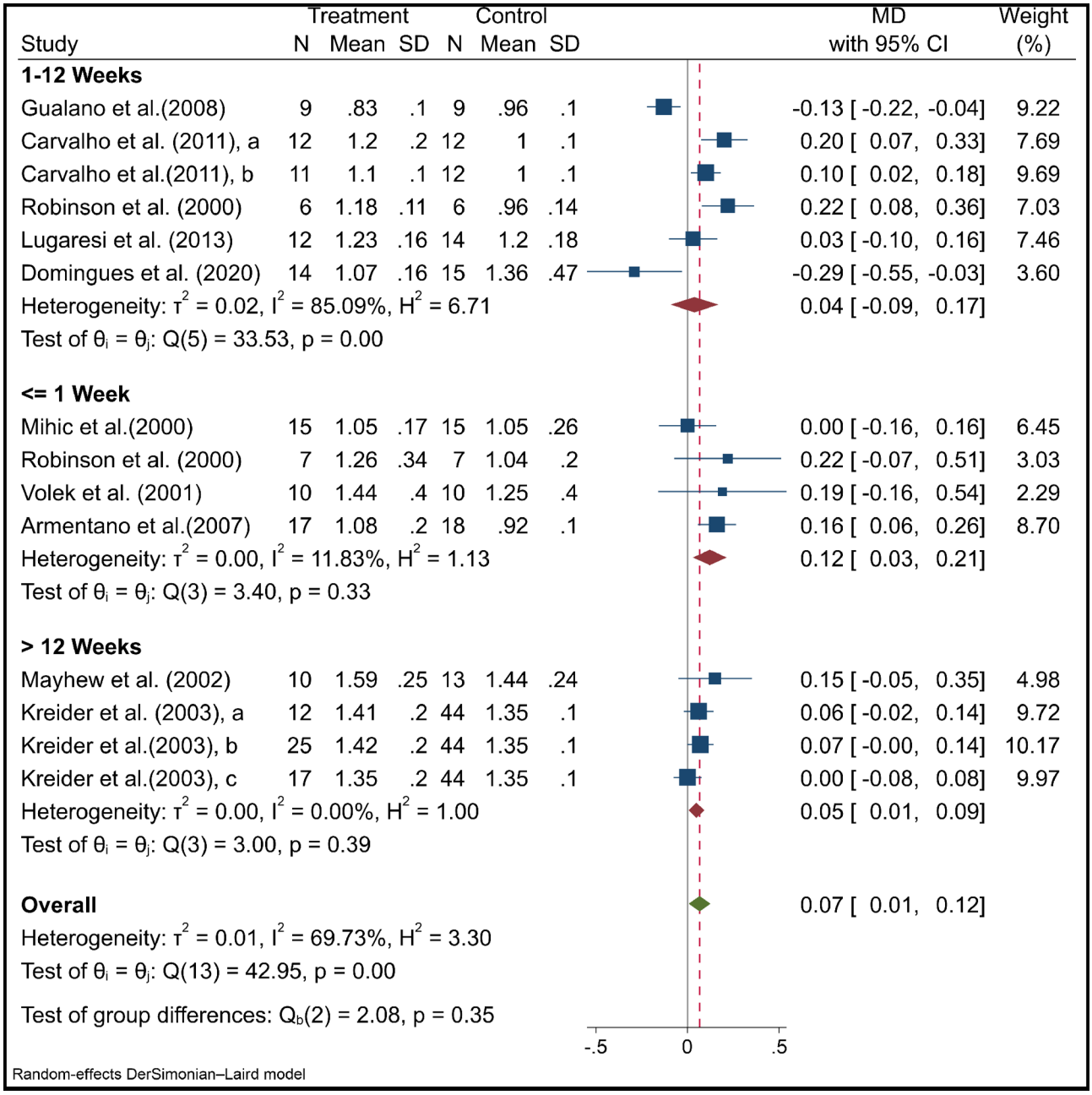


Fig. 3 Effects of creatine supplementation on serum creatinine levels in terms of follow-up duration

An important methodological consideration in interpreting our findings relates to the variability of creatinine assays across studies. In 2009, the global standard reference material SRM 967 was introduced, leading to substantial changes in calibration and reported creatinine values. Furthermore, different assay methodologies (e.g., compensated vs. uncompensated Jaffe methods, sarcosine oxidase–based assays, creatine imidohydrolase–based assays) vary in their susceptibility to interference by creatine itself. Such differences can result in systematic measurement biases and may partly explain the heterogeneity observed in our analyses. Unfortunately, the majority of included trials did not report the assay type or standardization method used, which precluded subgroup analysis according to assay methodology. This lack of reporting should be considered an important limitation. Future research on creatine supplementation and kidney function would greatly benefit from standardized reporting of assay type and calibration method, which would enable more precise synthesis of evidence.

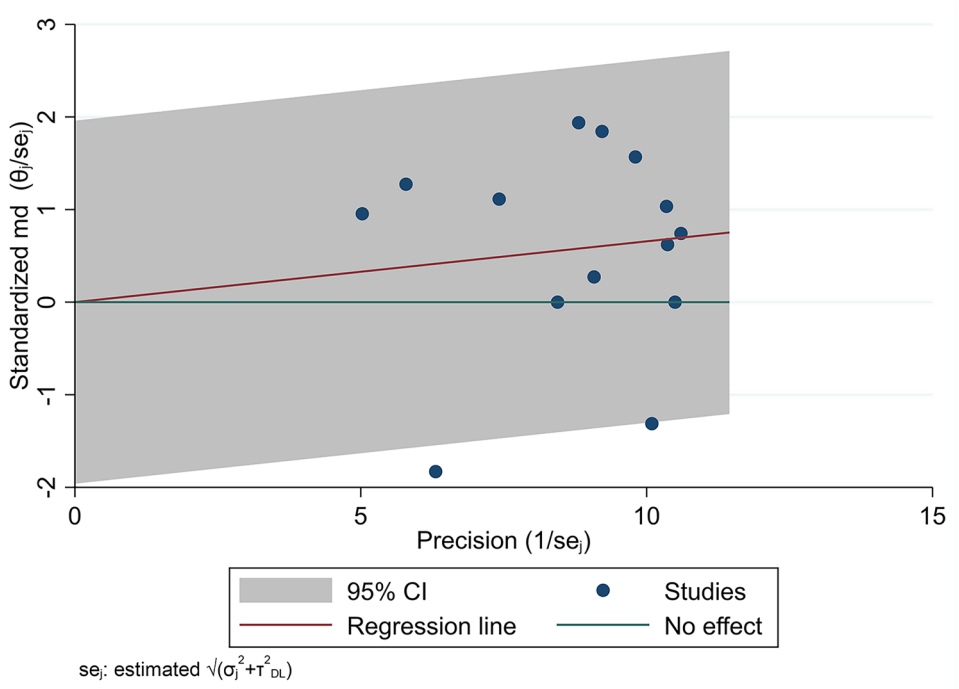


Fig. 4 Funnel plot of the SMDs plotted against study precision (1/SE)

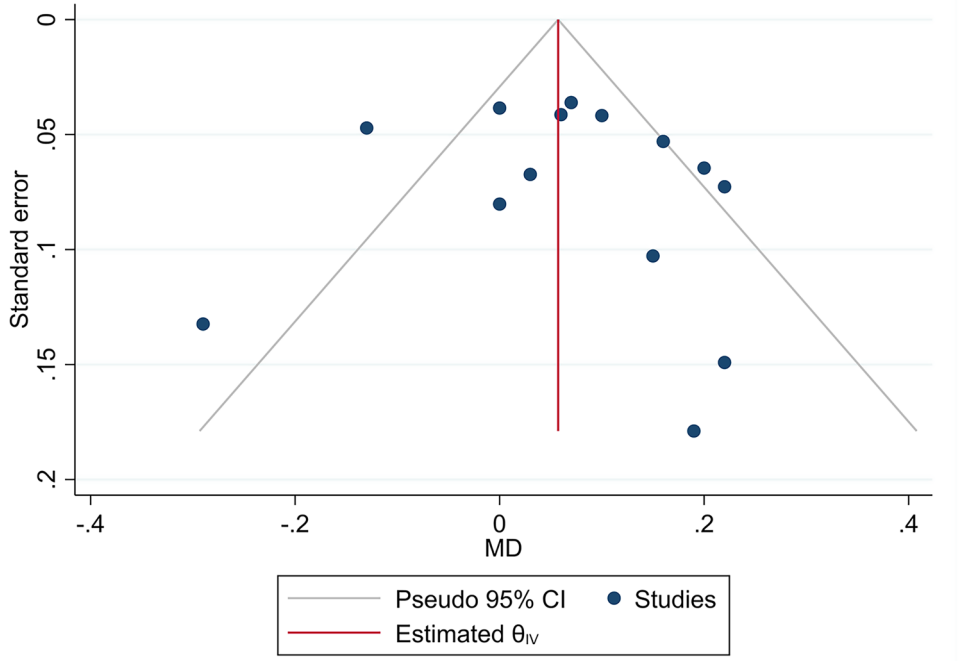


Fig. 5 Classic funnel plot based on mean differences

Limitations

Most included studies were of short to moderate duration; long-term safety data beyond one year remain sparse. In our study, we conducted subgroup analysis by intervention duration, as this was the most consistently reported variable across trials. However, information regarding dosage and participants' training status was either limited or inconsistently reported, which restricted our ability to perform further subgroup analyses. Future studies should provide detailed reporting of these variables to enable more comprehensive subgroup analyses.

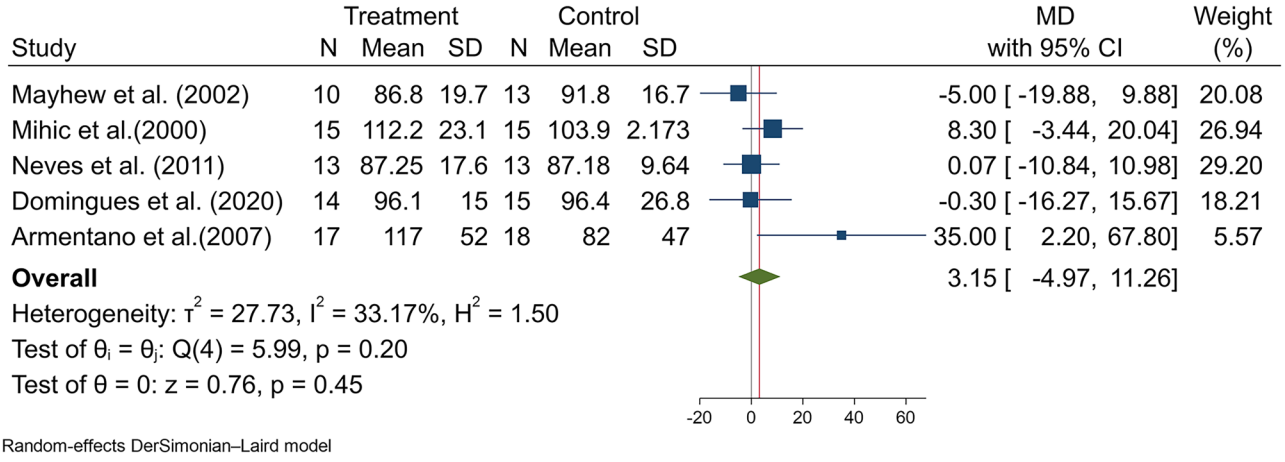


Fig. 6 Effects of creatine supplementation on GFR

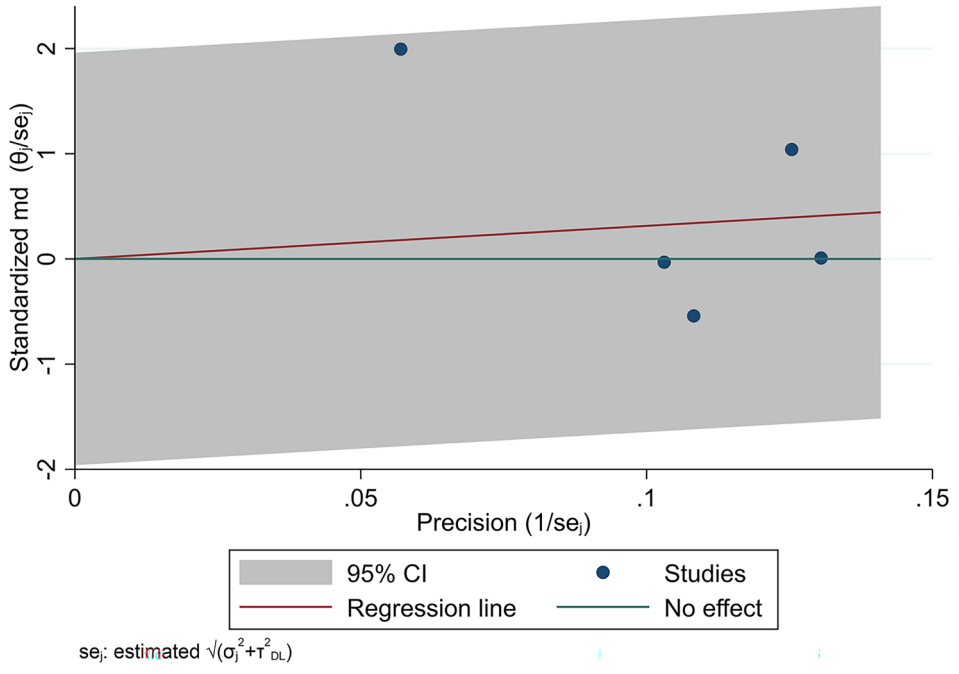


Fig. 7 Funnel plot of the SMDs plotted against study precision (1/SE) for GFR studies

Conclusion

In summary, this systematic review and meta-analysis demonstrate that creatine supplementation is associated with a modest increase in serum creatinine levels but does not adversely affect glomerular filtration rate. These results indicate that creatine is likely safe for kidney function in healthy individuals and various clinical populations when used within standard dosing protocols. However, researchers and clinicians should interpret elevated serum creatinine levels in the context of supplementation with caution and consider more specific renal function markers for accurate assessment.

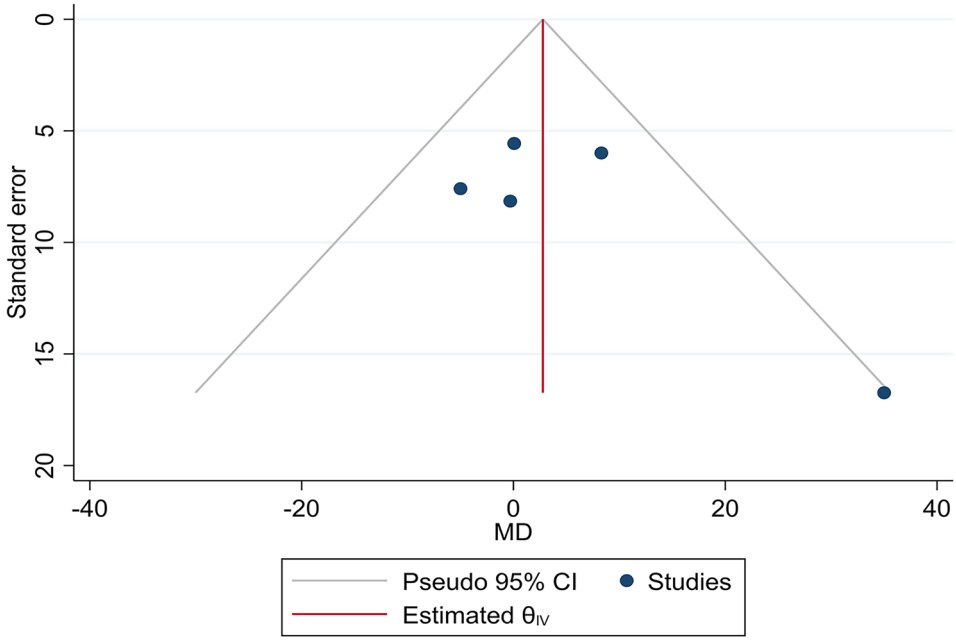


Fig. 8 Classic funnel plot based on mean differences for GFR studies

Table 2 Summary and details of risk bias assessment

Authors	Study design	Randomization/ Selection	Deviations/ Comparability	Missing data	Outcome measurement	Selective reporting	Overall risk
Gualano et al., 2008	RCT, double-blind	Some concerns	Low	Some concerns	Some concerns	Some concerns	Some concerns
Mayhew et al., 2002	Observational	Moderate	Moderate	Low/Moderate	-	-	Moderate risk
Carvalho et al., 2011	RCT, double-blind	Some concerns	Low	Some concerns	Low/Some concerns	Some concerns	Some concerns
Mihic et al., 2000	RCT, double-blind	Some concerns	Low	Low/Some concerns	Low	Some concerns	Some concerns
Robinson et al., 2000	RCT, placebo-controlled	Some concerns	Some concerns	Some concerns	Low	Some concerns	Some concerns
Volek et al., 2001	RCT, double-blind	Some concerns	Low	Some concerns	Low	Some concerns	Some concerns
Armentano et al., 2007	RCT, double-blind	Some concerns	Low	Some concerns	Low	Some concerns	Some concerns
Neves et al., 2011	RCT, placebo-controlled	Some concerns	Low	Some concerns	Low	Some concerns	Some concerns
Lugaresi et al., 2013	RCT, double-blind	Some concerns	Low	Some concerns	Low	Some concerns	Some concerns
Gualano et al., 2011	RCT, double-blind	Some concerns	Low	Some concerns	Some concerns	Some concerns	Some concerns
Domingues et al., 2020	RCT, double-blind	Some concerns	Low	Some concerns	Low	Some concerns	Some concerns
Kreider et al., 2003	Observational cohort	Moderate	Low/Moderate	Low/Moderate	-	-	Moderate risk

Abbreviations

ATP	Adenosine triphosphate
GFR	Glomerular filtration rate
Crn	Creatinine
Cr	Creatine
RCTs	Randomized controlled trials
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12882-025-04558-6.

Supplementary Material 1

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None.

Author contributions
E.K.N. and M.E. contributed equally to this work. They were primarily responsible for drafting the initial manuscript, conducting the literature search, performing data extraction and statistical analysis, interpreting the findings, and finalizing the manuscript for submission. M.M. , A.G. , and N.K. provided valuable support in data extraction, contributed to the data analysis process, and assisted in the interpretation of results. All authors reviewed and approved the final version of the manuscript.

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Data availability
The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
Not applicable because this is systematic review and meta-analysis.

Consent for publication
Not applicable because this is systematic review and meta-analysis.

Competing interests
The authors declare no competing interests.

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